Week 7. R Practice

PUBLISHED
April 16, 2026

```
library(ggplot2)
library(dplyr)
# install.packages("agricolae")
library(agricolae)
library(DescTools)
```

각 집단의 분포는 같지만 위치는 μ_i 만큼 다를 것이다.

(1) Permuational F-test

서로 다른 집단의 분포가 같은지 평균(μ_i)을 중심으로 검정.

$$F_i(x) = F(x - \mu_i),\ i = 1, \ldots, K$$

$$H_0 : F_1(x) = \ldots = F_K(x)$$

→ Under H_0, 집단간 분포 차이 X

Example: 주말 오후 1-3시에 서울 잠실 일대 공유 전동킥보드 세 브랜드(킥고잉, 라임, 고고씽)의 이용 시간을 조사하였다. K=3

```
kick <- read.csv("kickboard.csv")
head(kick)
```

```
  time brand
1   10    KG
2   10    KG
3   10    KG
4   10    KG
5   10    KG
6   11    KG
```

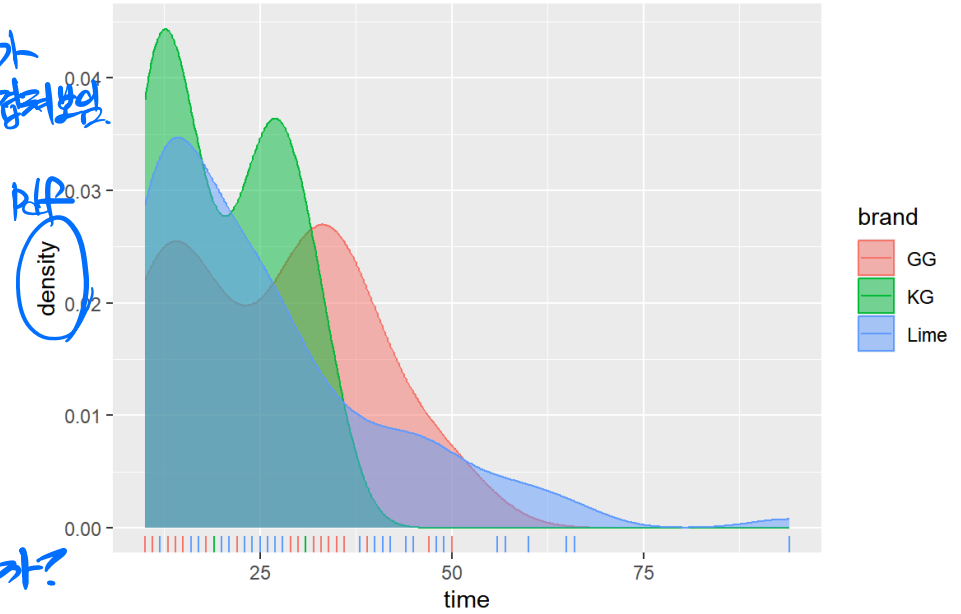
```
group_by(kick, brand) %>%
  summarise(count=n(), mean=mean(time), sd=sd(time))
```

```
# A tibble: 3 × 4
  brand count  mean    sd
  <chr> <int> <dbl> <dbl>
1 GG       22  26.2 12.5
2 KG       32  20.1  8.13
3 Lime    101  25.2 15.8
```

KG가 제 작아보임.
But, 평균을 비교했지만 표본분산도 각각 달라서 단순 평균으로는 비교 어려움

```
ggplot(kick, aes(x=time, color=brand, fill=brand)) +
  geom_density(alpha=0.5) + geom_rug()
```

분포가 꽤 겹쳐보임



그럼 어떻게 가설을 검정할까?

```
summary(aov(time ~ brand, data=kick))
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
brand        2    739     370    1.86  0.16
Residuals  152  30287     199
```

$F \approx 1.86$

$p \approx 0.16$

실제로 어떻게 구한 ANOVA 식 분해

```
N <- nrow(kick); K <- 3

group.length <- aggregate(kick$time, by=list(kick$brand), length)[,2]
group.mean <- aggregate(kick$time, by=list(kick$brand), mean)[,2]

SSTotal <- var(kick$time) * (length(kick$time)-1)
SST <- sum(group.length * (group.mean - mean(kick$time))^2)
SSE <- SSTotal - SST

F.obs <- (SST/(K-1))/(SSE/(N-K)); F.obs
```

```
[1] 1.8555
```

n_i : 각 집단 크기

$\bar{X}_i$: 각 집단 평균

$\bar{X}$: 전체 평균

$$SS_{Total} = \sum_{j=1}^{N} (X_j - \bar{X})^2$$

$$SST = \sum_{i=1}^{K} n_i (\bar{X}_i - \bar{X})^2$$

$$SSE = SS_{Total} - SST$$

$$F_{obs} = \frac{SST/(K-1)}{SSE/N-K}$$

```
R = 999 # number of permutations
F.perm <- vector(length=R)
for (r in 1:R) {
  perm <- sample(1:N, replace=FALSE)
  group.length <- aggregate(kick$time, by=list(kick$brand[perm]), length)[,2]
  group.mean <- aggregate(kick$time, by=list(kick$brand[perm]), mean)[,2]

  SSTotal <- var(kick$time) * (length(kick$time)-1)
  SST <- sum(group.length * (group.mean - mean(kick$time))^2)
  SSE <- SSTotal - SST
  F.perm[r] <- (SST/(K-1))/(SSE/(N-K))
}

p.value <- (sum(F.perm>=F.obs)+1)/(R+1); p.value
```

F_{perm} : permutation으로 얻은 F값

$$p\text{-value} = \frac{\#\{F_{perm} \geq F_{obs}\} + 1}{R+1}$$

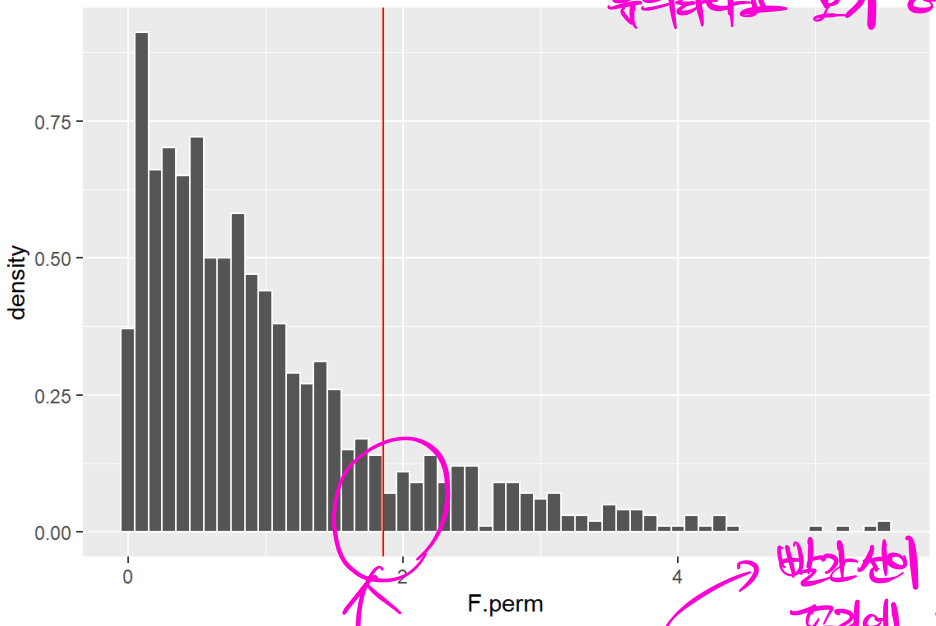
```
[1] 0.153
```

← p-value ⇒ ANOVA p-value와 매우 유사. ≈ >0.05

∴ Cannot reject H_0

```
data.frame(F.perm=F.perm) %>% ggplot(aes(x=F.perm)) +
  geom_histogram(aes(y=..density..), binwidth=0.1, color="white") +
  geom_vline(xintercept=F.obs, color='red')
```

퀴보드 브랜드 간 이용시간 차이가 유의하다고 보기 어렵다.



빨간선이 분포의 극단적인 꼬리에 있지X

↓

p-value ↑

≈ H_0 기각X이 시각적으로 확인 가능

F_{obs}의 위치 ≈ 1.86

비모수적으로 여러 집단 비교

(2) Kruskal-Wallis test: without ties

$$F_i = F(x - \mu_i),\ i = 1, \ldots, K$$

$$H_0 : F_1(x) = \ldots = F_K(x)$$

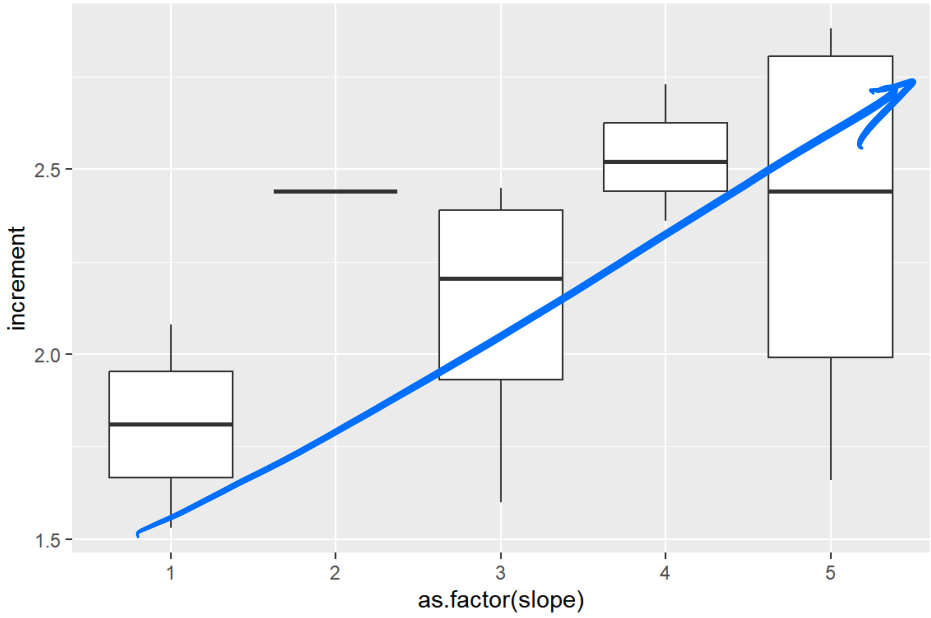
Example: 떡갈나무 16그루에 대해 연간 밑둥 직경의 증가량을 측정하였다. 각 나무는 위치한 부지의 경사도에 따라 분류되었다. (1: 경사 급함 ~ 5: 경사 없음)

```
tree <- read.csv("tree.csv")
head(tree)
```

key point: 작은값부터 큰값까지 순위를 매김

```
  increment slope
1      1.91     1
2      1.53     1
3      2.08     1
4      1.71     1
5      2.44     2
6      2.45     3
```

```
ggplot(tree, aes(x=as.factor(slope), y=increment)) + geom_boxplot()
```



slope↑ ⇒ increment↑ but sample↓
& sd varies ....
→ 아직 단정짓긴 어려움

```
kruskal.test(increment ~ slope, data=tree)
```

```
    Kruskal-Wallis rank sum test

data:  increment by slope
Kruskal-Wallis chi-squared = 5.97, df = 4, p-value = 0.2
```

```
tree.rank <- tree %>% mutate(rank=rank(increment))
head(tree.rank)
```

```
  increment slope rank
1      1.91     1    5
2      1.53     1    1
3      2.08     1    7
4      1.71     1    4
5      2.44     2   11
6      2.45     3   12
```

```
group.length <- aggregate(rank ~ slope, data=tree.rank, length)[,2]
group.rank.mean <- aggregate(rank ~ slope, data=tree.rank, mean)[,2]
```

group별 표본수 n_i
group별 평균순위 $\bar{R}_i$

```
N <- sum(group.length); K <- length(group.length)
KW.obs <- ( 12/(N*(N+1))) * sum( group.length * (group.rank.mean - (N+1)/2)^2 )
KW.obs
```

KW 통계량 직접 구하기

```
[1] 5.9669
```

$\chi^2_{5-1} = \chi^2_4$로 근사해서 p-value 계산.

```
p.value <- pchisq(KW.obs, df=K-1, lower.tail=F)
p.value
```

```
[1] 0.20163
```

> 0.05 ⇒ ∴ 경사도에 따라 증가량 분포가 다르다고 볼 충분한 증거가 없다.

비모수적으로 여러 집단 비교 ⇝ 단, 집단에 자연스러운 순서가 있을 때 사용.

(3) Jonckheere-Terpstra Test: ordered alternative

존키어 - 터프스트라

$H_a : F_i(x) \geq F_j(x)$ for $i < j$ ⟺ 경사도 수준이 커질수록 증가량이 커지는가?

```
tree$slope <- factor(tree$slope, ordered=TRUE)
# exact test
DescTools::JonckheereTerpstraTest(increment ~ slope, data=tree, alternative="increasing")
```

1 2 3

$F_1 \geq F_2 \geq F_3$

```
    Jonckheere-Terpstra test

data:  increment by slope
JT = 73, p-value = 0.015
alternative hypothesis: increasing
```

< 0.05 ∴ 경사도 ↑ ⟹ 증가량 ↑ 하는 순서구조가 있다.

※ KW : 집단들 중 어딘가 차이가 있나?
JT : 집단이 순서대로 증가하나?

(4) Multiple testing: controlling FWER

KW : H_0 : 모든 값의 분포가 같다
JT : H_a : 집단 순서에 따라 값이 증가한다 (= ordered alternative)

이 자료에선 더게 더 적절.

$X_{l,1}, \ldots, X_{1,100} \sim N(\mu_l, 1),\ l = 1, \ldots, 20$

$H_{0l} : \mu_l = 0,\ l =, \ldots, 20$

The following simulations shows that setting $\alpha = 0.05$ for each test leads to undesired results.

```
L=20 # number of tests
n=100 # size of samples
R=1000
err.vec <- vector(length=R)
for (r in 1:R) {
  fam.err <- FALSE
  x=matrix(rnorm(n*L), nrow=n)
  for (i in 1:L) {
    if (t.test(x[,i], mu=0)$p.value < 0.05) {
      fam.err <- TRUE
      break
    }
  }
  err.vec[r] <- fam.err
}

(fam.err.rate <- sum(err.vec)/R)
```

⇝ 여러개의 가설을 각각 유의수준 $\alpha = 0.05$로 따로 검정하면, 전체적으로 적어도 하나의 제1종 오류가 날 확률이 커진다.

```
[1] 0.612
```

Given that $H_{0l},\ l = 1, \ldots, 20$ being all true,

FWER = Pr(at least one type 1 error in the family)

= 1 - Pr(no type 1 error in the family)

= 1- (1 - 0.05)^{20}

```
1 - (1-0.05)^20
```

```
[1] 0.64151
```

→ 20보다 크면, FWER도 그만큼 증가 ∴ 개별 검정의 0.05는 전체 family 기준으로는 느슨하게 나섰을 가능할 수 있음

Need to use a smaller significance level $\alpha\prime$ for each pairwise test!

(5) Multiple comparisons: Bonferroni adjustment

FWER을 제어하기 위해 α를 더 낮게 잡는 방법

Example: Percentages of clay in soil samples from four locations. (Populations are normally distributed with equal variances.)

$\alpha' = \alpha/m$

, m : 비교의 갯수.

```
clay <- read.csv("clay.csv")
head(clay)
```

```
   pct location
1 26.5        1
2 15.0        1
3 18.2        1
4 19.5        1
5 23.1        1
6 17.3        1
```

```
group_by(clay, location) %>%
  summarise(count=n(), mean=mean(pct), sd=sd(pct))
```

```
# A tibble: 4 × 4
  location count  mean    sd
     <int> <int> <dbl> <dbl>
1        1     6  19.9  4.19
2        2     6  19.8  6.35
3        3     6  25.4  5.89
4        4     6  30.3  4.33
```

```
summary(aov(pct ~ as.factor(location), data=clay))
```

```
                    Df Sum Sq Mean Sq F value Pr(>F)
as.factor(location)  3    452   150.8    5.42 0.0068 **
Residuals           20    556    27.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Bonferroni adjustment is to do each of the $K(K-1)/2 = 6$ comparisons at level of significance $\alpha\prime = \alpha/6$.

```
pct.all <- matrix( c(clay$pct[clay$location==1],
                     clay$pct[clay$location==2],
                     clay$pct[clay$location==3],
                     clay$pct[clay$location==4]),
```

```
                    byrow=FALSE, ncol=4 )

pvals <- matrix(, nrow=4, ncol=4)
for (i in 1:4) {
  j <- i+1
  while(j<=4) {
    pvals[i,j] <- t.test(pct.all[,i], pct.all[,j], var.equal=TRUE)$p.value
    j <- j+1}
}

pvals # raw p-values
```

```
     [,1]    [,2]     [,3]      [,4]
[1,]   NA 0.97911 0.096298 0.0017560
[2,]   NA      NA 0.150854 0.0075116
[3,]   NA      NA       NA 0.1270221
[4,]   NA      NA       NA        NA
```

```
pvals[is.na(pvals)==FALSE] <- p.adjust(pvals[is.na(pvals)==FALSE], method="bonferroni")
```

```
pvals # Bonferroni-adjusted p-values
```

좀더 커짐 (보수적↑)

```
     [,1] [,2]    [,3]     [,4]
[1,]   NA    1 0.57779 0.010536
[2,]   NA   NA 0.90512 0.045070
[3,]   NA   NA      NA 0.762133
[4,]   NA   NA      NA       NA
```

```
t.test(clay$pct[clay$location==2], clay$pct[clay$location==4], var.equal=TRUE)$p.value*6
```

```
[1] 0.04507
```

```
(bonf.critical.value <- qt(0.025/6, df=2*6-2))
```

```
[1] -3.2768
```

(6) Multiple comparisons: Fisher's protected LSD

; 1st 전체 ANOVA가 유의한지 확인

Fisher's LSD procedure is to declare means for treatments i and j to be different if

$$|\bar{X}_i - \bar{X}_j| \geq t(\alpha/2, df = N - K)\sqrt{MSE(2/6)}$$

⇒ 2nd 두집단 평균차이를 비교

```
N <- nrow(clay); K <-4
clay$location <- as.factor(clay$location)
model <- aov(pct ~ location, data=clay)

MSE <- deviance(model)/df.residual(model)
```

⊕ ; Bonferroni보다 덜 보수적이라서 차이를 더 잘 잡음

⊖ error control 약해짐 (ex type 1)

```r
(lsd.critical.value <- qt(0.025, df=N-K) * sqrt(MSE*(2/6)))
```

```
[1] -6.3508
```

```r
(agricolae::LSD.test(model, "location", p.adj="none"))
```

```
$statistics
  MSerror Df   Mean     CV t.value    LSD
   27.808 20 23.863 22.099   2.086 6.3508

$parameters
        test p.ajusted   name.t ntr alpha
  Fisher-LSD      none location   4  0.05

$means
     pct    std r     se    LCL    UCL  Min  Max    Q25   Q50    Q75
1 19.933 4.1860 6 2.1528 15.443 24.424 15.0 26.5 17.525 18.85 22.200
2 19.850 6.3463 6 2.1528 15.359 24.341 14.1 30.2 15.975 16.95 23.175
3 25.350 5.8930 6 2.1528 20.859 29.841 19.2 35.0 21.450 23.80 28.175
4 30.317 4.3250 6 2.1528 25.826 34.807 26.3 37.3 27.025 29.05 32.650

$comparison
NULL

$groups
     pct groups
4 30.317      a
3 25.350     ab
1 19.933      b
2 19.850      b

attr(,"class")
[1] "group"
```

(7) Multiple comprisons: Tukey's HSD

모든 pair 평균 비교를 하되, 가장 큰 평균치를 기준으로 FWER 통제.

Tukey's HSD procedure considers a statistic that measures the largest difference between sample means:

⇒ pairwise 비교에서 FWER 가장 잘 통제.

$$Q = \max_{ij}\{\frac{\sqrt{n}|\bar{X}_i - \bar{X}_j|}{\sqrt{MSE}}\} = \frac{\sqrt{n}(\max_i \bar{X}_i - \min_i \bar{X}_i)}{\sqrt{MSE}}.$$

The procedure declares means for treatment i and j to be different if

$$|\bar{X}_i - \bar{X}_j| / \sqrt{MSE/n} \geq q(\alpha, K, df = N - k)$$

where $q(\alpha, K, df = N - k)$ is the upper-tail $100\alpha\%$ point of the distribution of Q.

```r
(out <- TukeyHSD(model, "location", ordered=FALSE))
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = pct ~ location, data = clay)

$location
         diff     lwr     upr   p adj
2-1 -0.083333 -8.6048  8.4381 0.99999
3-1  5.416667 -3.1048 13.9381 0.31194
4-1 10.383333  1.8619 18.9048 0.01357
3-2  5.500000 -3.0215 14.0215 0.29951
4-2 10.466667  1.9452 18.9881 0.01277
4-3  4.966667 -3.5548 13.4881 0.38453
```

```r
Tij.obs <- abs(out$location[,1]); Tij.obs
```

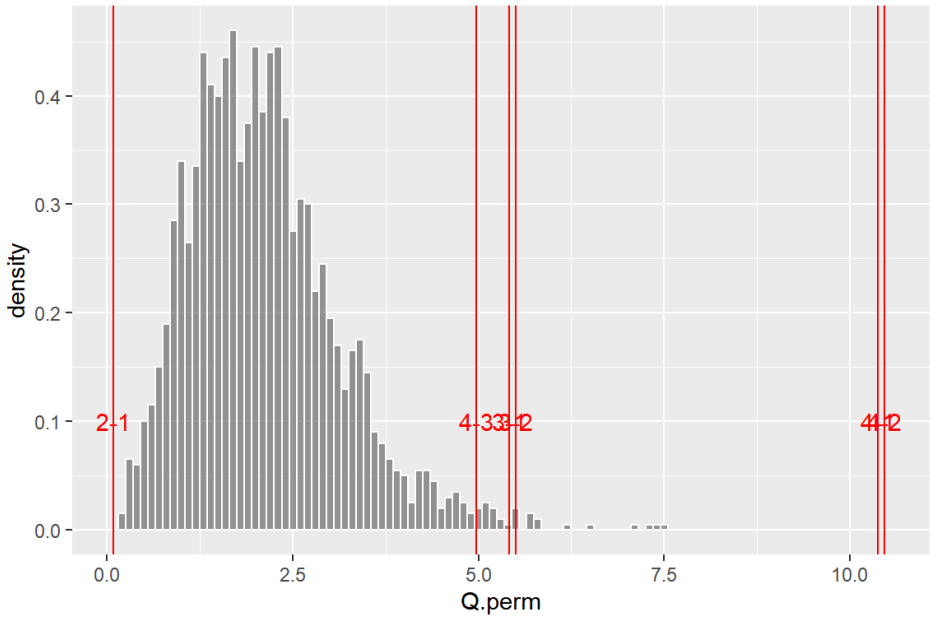
```
      2-1       3-1       4-1       3-2       4-2       4-3
 0.083333  5.416667 10.383333  5.500000 10.466667  4.966667
```

```r
R = 2000
Q.perm <- vector(length=R)
for (r in 1:R) {
  clay.perm <- data.frame(pct = clay$pct, location=clay$location[sample(1:N, replace=FALSE)])
  model <- aov(pct ~ location, data=clay.perm)
  MSE <- deviance(model)/df.residual(model)
  group.mean <- aggregate(pct ~ location, data=clay.perm, mean)[,2]
  Tij <- abs(outer(group.mean, group.mean, "-"))/ sqrt(MSE/6)
  Q.perm[r] <- max(Tij)
}
```

```r
( p.values <- unlist(purrr::map(Tij.obs, function(x) {sum(Q.perm>=x)/R})) )
```

```
   2-1    3-1    4-1    3-2    4-2    4-3
1.0000 0.0075 0.0000 0.0060 0.0000 0.0150
```

```r
data.frame(Q.perm=Q.perm) %>% ggplot(aes(Q.perm)) +
  geom_histogram(aes(y=..density..), binwidth=0.1, color="white", alpha=0.6) +
  geom_vline(xintercept=Tij.obs, color="red") +
  geom_text(data=data.frame(x=Tij.obs, label=names(Tij.obs)),
            aes(x=x, y=0.1, label=label), color="red")
```


density
0.0
0.1
0.2
0.3
0.4
2-1
4-3
3-2
4-2
0.0
2.5
5.0
7.5
10.0
Q.perm